

Rings and Modules

1. Find an example of each of the following.

- (a) A ring R and a left ideal $I \subset R$ that is *not* a right ideal.
- (b) A ring R and a prime ideal $P \subset R$ that is not maximal. (Let's make this harder and say $P \neq \{0\}$.)
- (c) A ring R , an ideal I , and a prime ideal $P \subset R/I$.
- (d) A ring R and an ideal P that is primary, but not prime.
- (e) A map of rings $f: R \rightarrow S$ and a prime ideal $P \subset R$ such that $f(P)S \subset S$ is *not* prime. (Can you find an example where $f(P)S \neq (1)$?)

1. Solution

- (a) Let $R = M_2(\mathbb{R})$, and let I be the set of matrices of the form

$$\begin{pmatrix} a & 0 \\ b & 0 \end{pmatrix}.$$

- (b) Let $R = \mathbb{C}[x, y]$, and let $P = (x)$.
- (c) Let $R = \mathbb{Z}$, $I = 6\mathbb{Z}$, and $P = 2\mathbb{Z}/6\mathbb{Z}$.
- (d) Let $R = \mathbb{C}[x]$, and $P = (x^2)$
- (e) Let $R = \mathbb{Z}$, $S = \mathbb{Z}[i]$, $P = (2)$.

2. (January 2020 Problem 3(a) (modified)) Let R be a commutative ring, I an ideal in R , and S a multiplicative subset of R . There are two standard correspondences involving prime ideals:

- (i) Prime ideals in R/I are in bijection with prime ideals in R which contain I .
- (ii) Prime ideals in $S^{-1}R$ are in bijection with prime ideals in R which do not meet S .

Prove (i) and (ii).

2. Solution

- (i) Let P be a prime ideal which contains I . Then P/I is a proper ideal of R/I . Let $x, y \in R$ such that $[x][y] \in P/I$, where $[x]$ denotes the class of x in R/I . Then $xy = p + i$ for some $p \in P$ and $i \in I$. Since $P \supset I$, we in fact have $xy \in P$, which means $x \in P$ or $y \in P$. Thus, $[x] \in P/I$ or $[y] \in P/I$.

On the other hand, let $Q \subset R/I$ be a prime ideal, and consider $P = \pi^{-1}(Q)$, where $\pi: R \rightarrow R/I$ is the quotient map. Then P is an ideal of R . Let $xy \in P$, then $[x][y] \in Q$, and so either $[x] \in Q$ or $[y] \in Q$. Thus, either $x \in P$ or $y \in P$.

- (ii) Let P be a prime ideal which does not meet S : $P \cap S = \emptyset$. Then $S^{-1}P$ is a proper ideal of $S^{-1}R$. Suppose $\frac{x}{s_1} \cdot \frac{y}{s_2} \in S^{-1}P$, then there exist $p \in P$ and $s_3 \in S$ such that $xy s_3 = p s_1 s_2$. Notice here that the RHS is in P , so $xy s_3 \in P$. But $s_3 \notin P$, so $xy \in P$, and thus either $x \in P$ or $y \in P$. Therefore, either $\frac{x}{s_1} \in S^{-1}P$ or $\frac{y}{s_2} \in S^{-1}P$.

On the other hand, let $Q \subset S^{-1}R$ be a prime ideal, and let $P = \ell^{-1}(Q)$, where $\ell: R \rightarrow S^{-1}R$ is the localization map. From here, the argument is identical to the above.

- 3.** Let R be a local ring with maximal ideal $\mathfrak{m}$. Show that $x \in R$ is a unit if and only if $x \notin \mathfrak{m}$.

3. Solution It is clear that if x is a unit, then $x \notin \mathfrak{m}$, since $\mathfrak{m} \neq (1)$. On the other hand, if x is not a unit, then (x) is a proper ideal, and hence is contained in a maximal ideal. But there's only one maximal ideal, so $x \in \mathfrak{m}$.

- 4.** Let R be a commutative ring, and S a multiplicative subset of R .

- (a) When is $S^{-1}R = 0$?
- (b) When is the localization map $R \rightarrow S^{-1}R$ injective?

4. Solution

- (a) $S^{-1}R = 0$ if and only if $0 \in S$
- (b) The localization map is injective if and only if there are no zero divisors in S .

5. (August 2019 Problem 2) This problem involves ideals in a commutative ring (with unit). A proper ideal I is *primary* if whenever $ab \in I$ and $a \notin I$, then $b^n \in I$ for some $n > 0$. The radical of an ideal I , denoted $\sqrt{I}$, is the set $\sqrt{I} = \{f : f^n \in I \text{ for some } n > 0\}$.

- (a) Prove that if I is primary, then $\sqrt{I}$ is prime.
- (b) Is the ideal $J = (x^2, xy) \subset \mathbb{Q}[x, y]$ primary? Be sure to carefully justify your answer.

5. Solution

- (a) Suppose $fg \in \sqrt{I}$, then by definition there exists some $n_1 > 0$ such that $(fg)^{n_1} = f^{n_1} g^{n_1} \in I$. Since I is primary, either $f^{n_1} \in I$ — in which case $f \in \sqrt{I}$ — or else there exists some n_2 such that $(g^{n_1})^{n_2} = g^{n_1 n_2} \in I$. In the latter case, $g \in \sqrt{I}$. Thus, $\sqrt{I}$ is prime.
- (b) The ideal J is not primary. To see this, note that $xy \in J$, but neither $x \in J$ nor $y^n \in J$ for any $n > 0$.

6. (January 2018 Problem 2, modified) Consider the ring $R = \mathbb{C}[x]/(x^3)$.

- (a) Give an example of a nonzero zero-divisor in R .
- (b) Show that the set of zero-divisors in R form an ideal.

- (c) Let S be a commutative ring. Prove that if the set of zero-divisors is an ideal I , then $I \subset S$ is a prime ideal.

6. Solution

- (a) The element x is a nonzero zero-divisor.
- (b) Notice that every element of the ideal (x) is a zero-divisor. On the other hand, notice that the prime ideals of R are prime ideals of $\mathbb{C}[x]$ which contain (x^3) . There is only one such prime ideal, namely (x) , so R is a local ring with maximal ideal (x) . Thus, any element not in (x) is invertible, and so isn't a zero-divisor. Thus, every zero-divisor lives in (x) .
- (c) Suppose ab is a zero-divisor, but a is not a zero-divisor. Then there exists some nonzero c such that $(ab)c = 0$. Since a is not a zero-divisor, and since $a(bc) = 0$, it must be that $bc = 0$. But then b is a zero-divisor. Thus, if the set of zero-divisors is an ideal, it is a prime ideal.

7. (January 2018 Problem 1)

- (a) Give an example of a commutative ring R and a nonzero element $f \in R$ where the localization $R_f = 0$.
- (b) Give an example of a commutative ring R and a nonzero element $f \in R$ where the localization map $R \rightarrow R_f$ is neither injective nor surjective.
- (c) Give an example of a *local* ring R and an element $f \in R$ where $R_f \neq 0$ but R_f is no longer a local ring.

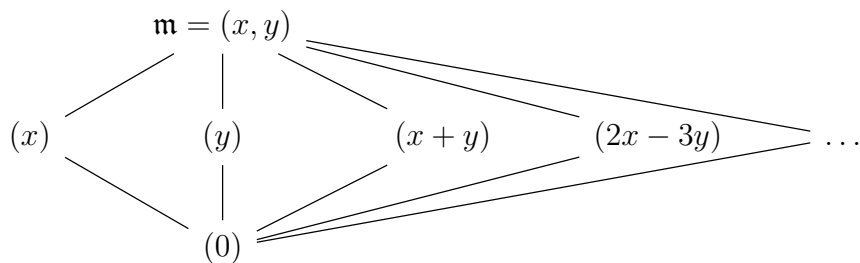
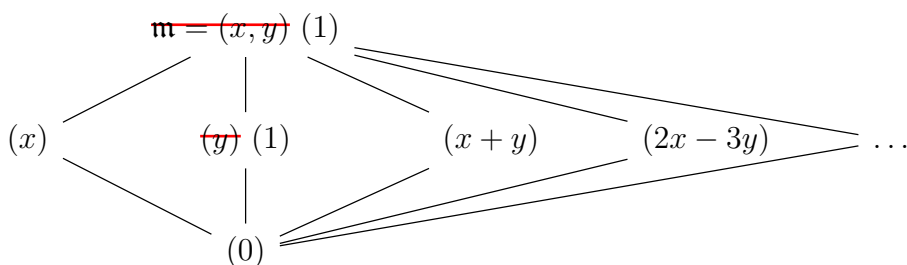
7. Solution

- (a) Let $R = \mathbb{Z}/4\mathbb{Z}$, $f = 2$
- (b) Let $R = \mathbb{C}[x, y]/(xy)$ and $f = x$. Then

$$\begin{aligned} R_f &= (\mathbb{C}[x, y]/(xy))_x \\ &= \mathbb{C}[x, y]_x/(xy)_x \\ &= \mathbb{C}[x, x^{-1}, y]/(xy) \\ &= \mathbb{C}[x, x^{-1}, y]/(y) \\ &= \mathbb{C}[x, x^{-1}]. \end{aligned}$$

Here, we can see this is not injective since $y \in R$ maps to 0, but it is also not surjective because nothing maps to x^{-1} .

- (c) Let $R = \mathbb{C}[x, y]_{(x, y)}$ and $f = y$. Remember the notation, R is of the form $\mathbb{C}[x, y]_P$ for a prime ideal P . Let's look at the structure of the prime ideals in R , also known as its **Spectrum**. The ring R is local, and has unique maximal ideal $\mathfrak{m} = (x, y)$, but it has many other prime ideals contained in $\mathfrak{m}$, such as the ideals (x) , (y) , $(x + y)$, $(2x - 3y)$, and so on. When we localize at $f = y$, all the primes which contain y will become the


 Figure 1: The spectrum of R from 7(c)

 Figure 2: The spectrum of R_f from 7(c)

unit ideal, but any prime which does not contain y will remain prime in the localization. Thus, the spectrum of R_f is pictured in Figure 2. We can see that the primes like (x) or $(x + y)$ become maximal in the localization, so that R_f is no longer a local ring.

8. (January 2020 Problem 3(b)-(c))

- Give an example of a ring with finitely many elements that has exactly three prime ideals.
- Give an example of a subring of $\mathbb{Q}$ that has exactly three prime ideals.

8. Solution

- $\mathbb{Z}/30\mathbb{Z}$
- Let $S = \{n \in \mathbb{Z} : 2 \nmid n, 3 \nmid n\}$. Then $S^{-1}\mathbb{Z} \subset \mathbb{Q}$ has only the prime ideals (0) , (2) , and (3) . (Note on notation: usually in this context people would actually write $\mathbb{Z}[\frac{1}{S}]$. This is to avoid the issues from the notational note on the original worksheet.)

9. Consider $M_2(\mathbb{Z})$, the ring of 2×2 matrices with entries in $\mathbb{Z}$.

- (Aug 2019 1(b) and Jan 2021 1(b))** Find a 2-sided ideal of $M_2(\mathbb{Z})$ that is neither 0 nor the whole ring.
- (Jan 2018 2(b), modified)** Give an example of a nonzero left zero-divisor of $M_2(\mathbb{Z})$.
- (Jan 2018 2(c), modified)** Show that the left zero-divisors of $M_2(\mathbb{Z})$ do not form a left ideal.

- (d) Find all the prime ideals of $M_2(\mathbb{Z})$. (For a noncommutative ring, a proper 2-sided ideal P is prime if whenever A, B are 2-sided ideals such that $AB \subset P$, then either $A \subset P$ or $B \subset P$.)

9. Solution

- (a) Consider the set of matrices with all even entries.
 (b) Consider the matrix

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

- (c) Notice that both the matrix above and

$$\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

are left zero-divisors, but their sum is the identity, which is not a zero-divisor.

- (d) We will show the prime ideals of $M_2(\mathbb{Z})$ are exactly $M_2(p\mathbb{Z})$ for p a prime.

Suppose I is an ideal of $M_2(\mathbb{Z})$. Suppose $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in I$. Then, multiplying by appropriate matrices we see that

$$\begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & a \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ a & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ 0 & a \end{pmatrix}$$

are all in I , and similarly for b, c, d . From this, it is easy to show that the set $\{a : a \text{ is an entry of an element of } I\}$ is an ideal of $\mathbb{Z}$. Hence, $I = M_2(n\mathbb{Z})$ for some n .

The final thing we need is that $M_2(n\mathbb{Z})M_2(m\mathbb{Z}) = M_2(nm\mathbb{Z})$, so the ideal is prime if and only if $I = M_2(p\mathbb{Z})$.